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Field studies of comfort and discomfort in sitting

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A previous study defined sitting comfort and discomfort as independent entities associated with different factors: discomfort is related to biomechanics and fatigue factors, and comfort to a sense of well-being and aesthetics. In this study a checklist for evaluation of chair comfort and discomfort was analysed in two field studies. In the first study two groups of subjects, ten secretaries and ten managers, evaluated two groups of ten chairs. Subjects assessed each chair three times during a workday using three different types of scales. The results of a factor analysis reconfirmed the factor structure of comfort and discomfort. Analysis of variance demonstrated that discomfort was related to fatigue accumulated during the workday, but it was not related to chair design. There was no significant Chair $\times$ Time period interaction, which implies that the rank order of preference among a set of chairs was established during the first assessment and did not change during the day. In a second field study 37 secretaries used three different formats of a Chair Evaluation Checklist with 14 items. The results of a factor analysis again confirmed the factor structure of comfort and discomfort. Analyses of variance demonstrated that subjects can evaluate comfort and discomfort simultaneously without any halo-effect. The results have methodological implications for measurement of comfort and discomfort. The findings for comfort, as defined, carry an important message that aesthetic design matters. This could provide a unifying focus for ergonomists and designers.

1. Introduction

Comfort and discomfort are frequently measured in ergonomic studies, especially in evaluations of workstations and chairs (Shackel *et al.* 1969, Helander *et al.* 1987). However, comfort is not a well-defined concept. In common parlance, it can refer to feelings of either comfort or discomfort and in ergonomic investigations scales often mix the two concepts. From a scientific perspective this may not be satisfactory. There are in fact several studies of sitting comfort that indicate that comfort and discomfort are affected by distinctly different variables (Habsburg and Middendorf 1978, Kleemam 1981, Kamijo *et al.* 1982), and assessment of comfort and discomfort should hence be based on different types of criteria (Helander *et al.* 1987, Kleeman 1981).

In a previous study by Zhang *et al.* (1996), multidimensional properties of comfort and discomfort were identified. Feelings of discomfort were associated with pain, tiredness, soreness and numbness. These feelings were assumed to be imposed by physical constraints in the design of the workstation, and mediated by factors

such as joint angles (Keegan 1959, Mandal 1984, Grandjean 1984, Michel and Helander 1994), tissue pressure (Hertzberg 1972, Kamijo *et al.* 1982), muscle contractions (Lundervold 1951), and blood pooling and circulation blockage (Winkel 1986). There is one very important observation: discomfort increases with time during the workday (Michel and Helander 1994). Assessment of comfort, on the other hand, was based on feelings of 'well-being' and the 'aesthetic' impression of the chair.

Since discomfort and comfort are based on independent factors a reduction of discomfort does not necessarily bring about feelings of comfort. In fact, from their independence it would follow that there is no connection between the two entities. This proposition may not seem totally reasonable, since one would assume that for comfort to occur the level of discomfort would have to be low. Conversely, great discomfort could only be experienced if the level of comfort is low. With the findings of the previous classification study, the authors proposed a conceptual model of sitting comfort and discomfort, which is conceptualized in figure 1.

The primary objectives of this study were to (1) develop a multidimensional scale for assessment of chair comfort and discomfort, and (2) validate the scale through field studies using office workers as test subjects. As part of the validation the results were compared to two existing methods for assessment of discomfort and comfort. They are described below.

In a previous study Helander *et al.* (1987) found that persons of small size often dislike large chairs, because the seat pad is too long and the lumbar support too high. Conversely, large size individuals do not like small chairs, because the seat pan is too short and the lumbar support is too low. A third objective was therefore to analyse the effect of subject stature on ratings of small and large chairs.

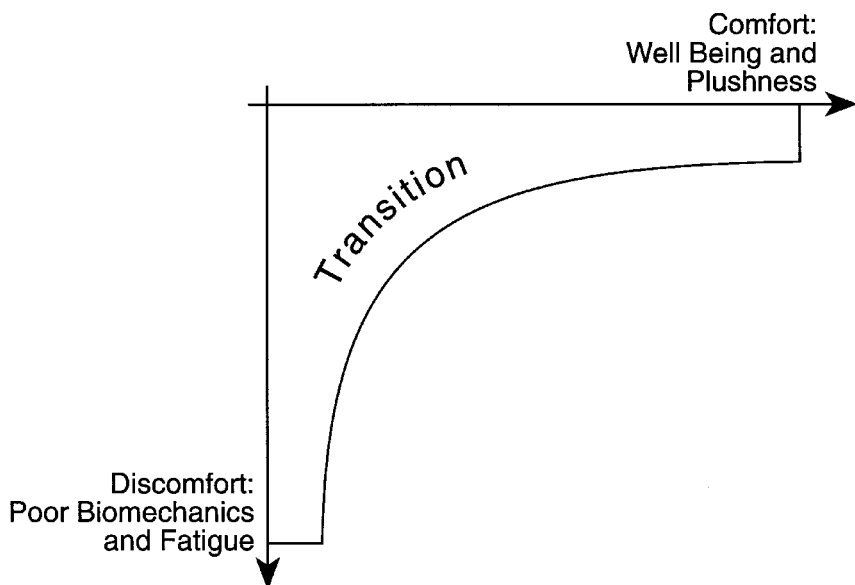


Figure 1. A conceptual model of sitting comfort and discomfort.

2. Field evaluation of comfort and discomfort

There are two well-established scales for assessment of comfort/discomfort. Shackel *et al.* (1969) developed the General Comfort Rating (GCR) scale for assessment of chair comfort/discomfort. This is a unidimensional 11-point scale with verbal anchors (table 1).

Corlett and Bishop (1976) published the Body Part Discomfort (BPD) Scale, which has general applicability to all types of work. To measure discomfort subjects are presented with an illustration of a human body that is segmented into several parts (shoulder, arm, etc.). In using the scale, subjects are guided by an experimenter to first consider which body part feels the most uncomfortable, and the discomfort is rated using a 5-point scale. The second most uncomfortable body part is then identified and rated, and so on. The sum of the discomfort ratings of all body parts constitute the BPD index, and this was the measure used in the present study.

Obviously, the design of GCR does not assume two independent factors. BPD focuses solely on discomfort, while in this study comfort and discomfort are emphasized as different and complementary constructs.

To construct the Chair Evaluation Checklist, 16 descriptors were selected from a previous study (Zhang *et al.* 1996), see figure 2.

The descriptors were chosen so as to be representative of the different dimensions of comfort and discomfort that had been identified. Compared to the previous study, a few of the descriptors were modified to better describe their intended meaning. Thus 'the chair looks nice' was used to represent aesthetic factors ('plushness' and 'luxurious'). 'I feel uneven pressure from seat pan or back' was used to represent feelings of not being supported by the chair. 'I like the chair' was added, since Habsburg and Middendorf (1978) claimed that this personal statement is the most important and most discriminating in chair evaluation. To assess the intensity of each descriptor, a 9-point Lickert scale was used.

3. Study 1

3.1. Method

3.1.1. *Chairs:* Twenty-one office chairs were selected from the current office furniture market by IBM Corporation. The particular chairs were selected because they passed certain criteria for sturdiness, durability and compliance with standards. Their ergonomic features were evaluated, and the chairs were ranked in order of preference

Table 1. The GCR scale for rating of chair comfort/discomfort.

Rating	Description
1	I feel completely relaxed
2	I feel perfectly comfortable
3	I feel quite comfortable
4	I feel barely comfortable
5	I feel uncomfortable
6	I feel restless and fidgety
7	I feel cramped
8	I feel stiff
9	I feel numb (or pins and needles)
10	I feel sore and tender
11	I feel unbearable pain

1. I have sore muscles	10. I feel relaxed
Not at all Moderately Extremely	Not at all Moderately Extremely
1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9
2. The chair looks nice	11. I feel numb
Not at all Moderately Extremely	Not at all Moderately Extremely
1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9
3. I have heavy legs	12. I feel uneven pressure
Not at all Moderately Extremely	Not at all Moderately Extremely
1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9
4. I feel stiff	13. The chair is spacious
Not at all Moderately Extremely	Not at all Moderately Extremely
1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9
5. I like the chair	14. I feel cramped
Not at all Moderately Extremely	Not at all Moderately Extremely
1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9
6. I feel restless	15. I feel refreshed
Not at all Moderately Extremely	Not at all Moderately Extremely
1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9
7. The chair feels soft	16. I feel restless
Not at all Moderately Extremely	Not at all Moderately Extremely
1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9
8. I feel tired	17. I feel comfortable (or "uncomfortable" in type B)
Not at all Moderately Extremely	Not at all Moderately Extremely
1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9
9. I feel pain	
Not at all Moderately Extremely	
1 2 3 4 5 6 7 8 9	

Figure 2. The first Chair Evaluation Checklist. *Instructions to subjects:* Below is a list of statements for evaluating this chair. There are several scales for you to rate. Mark an 'X' on each line at the point that best describes your feelings or your impressions. Note: 1 = Not at all; 9 = Extremely.

for possible inclusion in IBMs office furniture standard. These results have been reported elsewhere (Helander *et al.* 1995). In this study the authors were not interested in the comfort and ranking of individual chairs. The purpose was to validate and extend a previous study (Zhang *et al.* 1996). The chairs provided a convenient set of stimulus material for the present study.

Eleven of the chairs were manager's chairs and ten were secretaries' chairs. The managers' chairs were designed for males and were slightly larger in size than the secretaries' chairs, which were designed for females.

3.1.2 Subjects: Ten secretaries and ten managers from the administrative departments of SUNY at Buffalo participated as test subjects. This and the following study were both approved by the review board at the University. The subjects gave their informed consent for participating in the study. The demographics of the groups including stature are given in table 2. The secretaries evaluated the ten secretaries' chairs, and the managers evaluated the eleven managers' chairs. They were paid \$20 for participating in this study.

3.1.3. Procedure: It was suspected that simultaneous ratings of both comfort and discomfort in the same checklist could introduce a bias, since subjects might simply rate comfort the opposite of discomfort without further reflection. Two versions of the Chair Evaluation Checklist were therefore used, the only difference being the last item on the checklists. In version A the last questionnaire item asked for a rating of comfort, while version B evaluated discomfort.

Ten subjects (5 secretaries and 5 managers) used version A and the other ten subjects used version B. Subjects in both groups used each chair for a full 8 h workday. The chairs were randomly distributed among subjects on the evening before the day of testing. At the beginning of each day, the subjects were fully instructed how to use adjustability features on each chair. During each day the subjects filled out the forms for General Comfort Rating (GCR), Body Part Discomfort Scale (BPD), and Chair Evaluation Checklist (CEC) at three times: 09:00 h, 11:30 h and 16:30 h.

3.1.4. Evaluation of data: Altogether there were 210 subject-days of data collection with 60 questions asked each day for a total of 12,600 data points. Three analyses were performed:

- (1) Factor analysis to validate the multi-dimensional nature of comfort/discomfort.

Table 2. Demographic information.

	Secretarial†			Managerial‡		
	Range	Mean	SD	Range	Mean	SD
Age (years)	20–60	40	10·2	30–60	44	9·14
Stature (cm)	156–175	163	6	157–183	170	9
Weight (kg)	55–90	67	10	50–100	68	15

† 9 females, 1 male.

‡ 5 females, 5 males.

- (2) Analyses of variance to understand how the different items on CEC were influenced by three factors: chairs, stature, and time on task.
- (3) Pearson correlation coefficients were calculated for GCR, BPD, and the descriptors in CEC. This analysis was performed to understand differences between the three methods for measuring comfort/discomfort.

3.2. Results

3.2.1. *Factor analysis:* To verify the multidimensional nature of comfort/discomfort, ratings of the descriptors in CEC were submitted to factor analysis with *varimax* rotation (table 3). The first factor contained all major comfort descriptors, the second factor the discomfort descriptors, and the third factor grouped two descriptors related to chair size.

The factor loadings of factors 1 and 2 were plotted against the actual ratings of comfort and discomfort (figures 3 and 4).

3.2.2. *Analyses of variance:* Analyses of variance of the items in the CEC (figure 2) were performed for the secretaries. Significance levels of statistically significant results are given in table 4. There were three factors: chairs with ten levels, time periods for measurement with three levels, and subject stature with two levels. The same analyses were performed for managers, except that in this case there were 11 chairs (table 5).

The significant Chair XStature interaction for managers in table 5 were plotted. They were all similar implying that chair S, and to lesser extent chairs T and H produced ratings that were not uniform with the other chairs. Two of the interactions are displayed in figure 5.

3.2.3. *Correlation coefficients:* Pearson correlation coefficients were computed between ratings of the descriptors used in the CEC, GCR and BPD (table 6).

Table 3. Results of factor analysis of 16 descriptors in figure 2.

Descriptor	Factor 1: Comfort	Factor 2: Discomfort	Factor 3: Chair size
I feel restful	0.860	−0.179	−0.097
I feel refreshed	0.843	−0.183	−0.146
Chair looks nice	0.793	−0.044	−0.122
I like the chair	0.781	−0.261	−0.202
I feel relaxed	0.768	−0.256	0.131
Chair feels soft	0.656	−0.050	−0.225
I feel pain	−0.111	0.777	0.115
I have sore muscles	−0.146	0.760	−0.086
I feel restless	−0.325	0.734	0.080
I feel stiff	−0.341	0.720	−0.045
I feel tired	−0.025	0.584	0.167
I feel numb	−0.063	0.580	0.050
I have heavy legs	−0.052	0.543	−0.125
I feel uneven pressure	−0.352	0.533	0.259
I feel cramped	−0.186	0.186	0.903
Chair is spacious	0.567	0.191	−0.615
Extracted variance	4.458	3.775	1.493

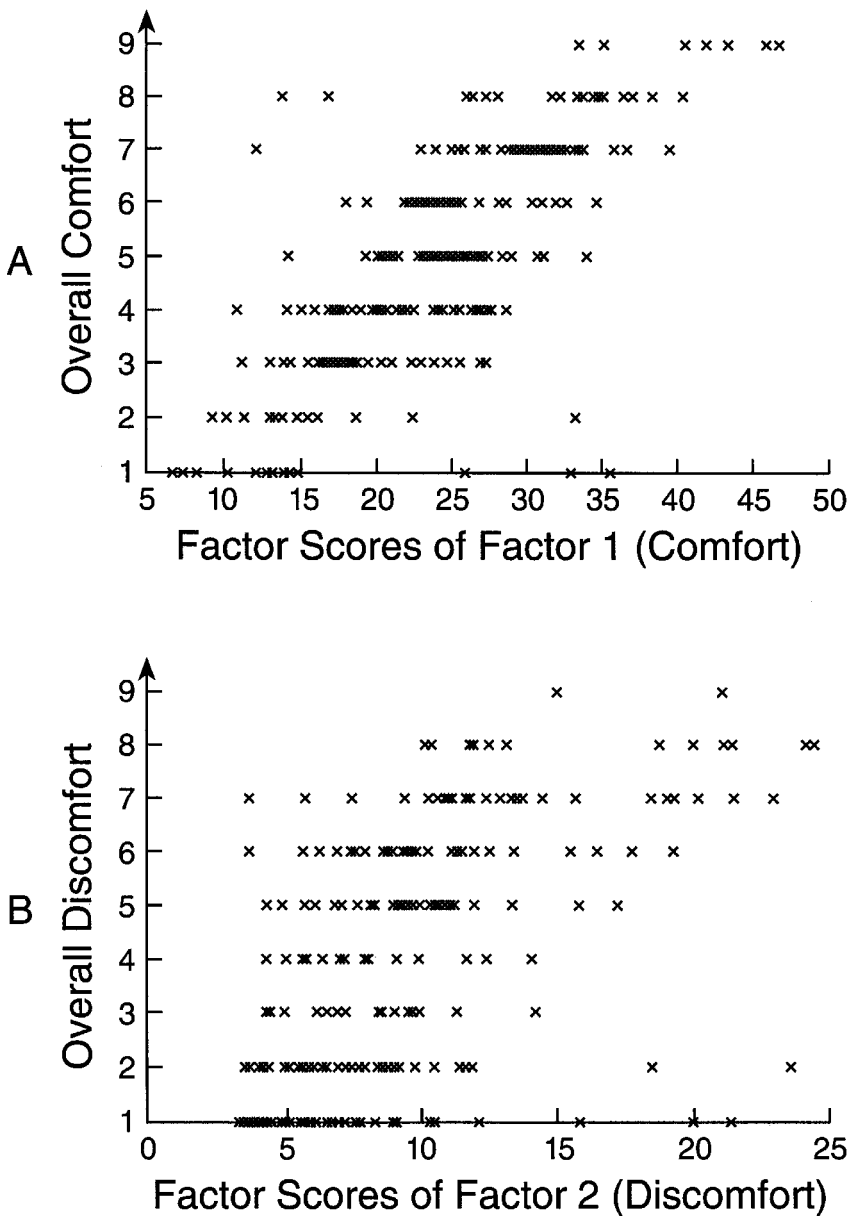


Figure 3. (A) Factor scores of Factor 1 (Comfort) plotted against ratings of overall comfort. (B) Factor scores of Factor 2 (Discomfort) plotted against ratings of overall discomfort.

3.3. Discussion

3.3.1. *Factor analysis:* The three factors in table 3 explained 61% of the variance. The first factor was interpreted as 'comfort', the second factor as 'discomfort', and the third factor as 'chair size'. These interpretations are in agreement with a previous study (Zhang *et al.* 1996), except that 'chair size' formed a third factor. Note,

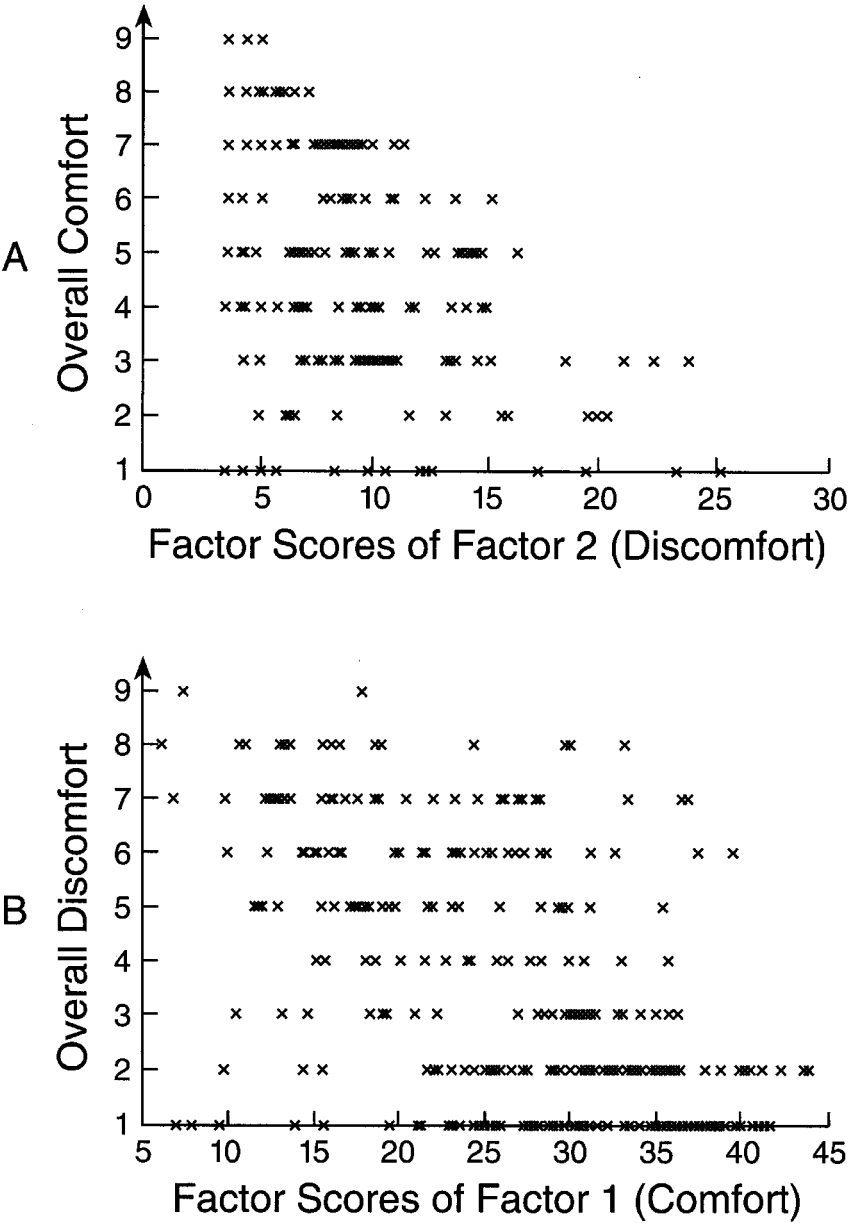


Figure 4. (A) Factor scores of Factor 2 (Discomfort) plotted against ratings of overall comfort. (B) Factor scores of Factor 1 (Comfort) plotted against ratings of overall discomfort.

however, that the last item ‘chair is spacious’, has a high factor loading on Factor 1, which is in agreement with our concept of comfort.

Figures 3 and 4 illustrate factor loadings plotted against overall ratings of ‘comfort’ and ‘discomfort’. Figure 3A demonstrates that factor scores of comfort were strongly correlated to overall ratings of comfort. Similarly factor

Table 4. Summary of results of ANOVAs of discomfort and comfort descriptors in the CEC. Ten secretaries evaluated 10 chairs for 1 day each. Significant results ($p < 0.05$) are given.

Descriptors	Chairs	Stature	Time period	C $\times$ S	C $\times$ T	S $\times$ T	C $\times$ S $\times$ T
<i>Discomfort descriptors</i>							
1. I have sore muscles			0.029				
2. I have heavy legs		0.0001					
3. I feel stiff	0.005		0.039				
4. I feel tired		0.001					
5. I feel pain			0.013				
6. I feel numb							
7. I feel uneven pressure	0.001	0.001					
8. I feel cramped	0.0001	0.047					
9. I feel restless	0.028						
<i>Comfort descriptors</i>							
10. I feel relaxed	0.0001						
11. I feel refreshed	0.0001						
12. I feel restful	0.0001	0.004					
13. Chair is spacious	0.0001	0.004					
14. Chair looks nice	0.0001	0.016					
15. Chair feels soft	0.0001						
16. I like the chair	0.0001	0.017					
<i>Overall evaluation</i>							
17. I feel comfortable	0.024	0.0001		0.0002			
18. I feel uncomfortable	0.036	0.001					

scores of descriptors of discomfort in figure 3B were correlated with overall discomfort.

The combinations of discomfort factor scores with overall comfort in figure 4A and comfort factor scores with overall discomfort in figure 4B indicate that there was fairly symmetrical transfer. In figure 4A, low values of discomfort factor scores were associated with a full range of values of overall comfort ratings from 1 to 9. However, the comfort ratings decreased sharply with increasing discomfort scores. In figure 4B the relationship between discomfort ratings and comfort factor scores was rather nebulous. The results presented may indicate that discomfort (physical strain) has a dominant effect. When discomfort factors are present, comfort factors become secondary in the perception of comfort/discomfort.

From a low level of discomfort it is not possible to predict overall comfort; it could be high or low. Similarly from a low level of comfort it is difficult to predict overall discomfort. However, from high values of either comfort or discomfort one can predict a low value of the other entity. These findings support the model of comfort/discomfort in figure 1.

3.3.2. Analyses of variance: The analyses of variance in tables 4 and 5 produced fairly consistent results for secretaries and managers. For the managers there were several significant Chair $\times$ Stature interactions. These were plotted, and they demonstrated that shorter subjects generally gave lower ratings for discomfort

Table 5. Summary of results of ANOVAs of discomfort and comfort descriptors in the CEC. Ten managers evaluated 11 chairs for one day . Significant results ($p<0.05$) are given.

Descriptors	Chairs	Stature	Time period	C ×S	C ×T	S ×T	C ×S ×T
<i>Discomfort descriptors</i>							
1. I have sore muscles			0.007				
2. I have heavy legs		0.013	0.040				
3. I feel stiff		0.001					
4. I feel tired		0.009	0.010				
5. I feel pain	0.032		0.001	0.002			
6. I feel numb							
7. I feel uneven pressure	0.003	0.0001					
8. I feel cramped	0.0001	0.003		0.0001			
9. I feel restless	0.008		0.001	0.014			
<i>Comfort descriptors</i>							
10. I feel relaxed	0.002		0.004				
11. I feel refreshed	0.002	0.003					
12. I feel restful	0.003	0.009					
13. Chair is spacious	0.0001	0.0001		0.0001			
14. Chair looks nice	0.0001	0.0001		0.0001			
15. Chair feels soft	0.0001	0.0001					
16. I like the chair	0.0001			0.0001			
<i>Overall evaluation</i>							
17. I feel comfortable	0.001			0.002		0.037	
18. I feel uncomfortable	0.0001	0.0001		0.0001			

descriptors, except for chair S which was judged to be higher than any other chair on the scales for ‘cramped’, ‘pain’, and ‘restless’. Taller subjects gave this chair average ratings. Similarly, shorter subjects rated the descriptors: ‘spaciousness’, ‘looks nice’, and ‘I like the chair’ relatively lower than taller subjects for chairs S and T. As representative samples the significant interaction for ‘pain’, and ‘spaciousness’ are given in figure 5.

Chair S was very special. It was equipped with concave rotating armrests that made it possible to rotate the lower part of the arms in the horizontal plane while resting the arms on the armrest. This design feature is intended to help individuals who suffer from repetitive motion injury. Obviously shorter subjects experienced this design feature more as a nuisance than a help; maybe because the armrests were too long, and their free rotation interfered with work activities. It is not meaningful to analyse the main effect of stature for the managers. Because of the significant interaction and the differences in response between short and tall subjects. For secretaries there were four significant main effects of stature for the discomfort descriptors. Tall secretaries had significantly greater discomfort than short secretaries due to ‘heavy legs’ (average value 3.17 versus 1.88) and ‘uneven pressure from seat back and seat pan’ (average values 3.81 versus 2.81). The latter item verifies the hypothesis that tall persons do not fit so well in small chairs. (The hypothesized results were also obtained for short managers, who gave ‘uneven pressure’ a greater rating, maybe because they did not fit well into large chairs.)

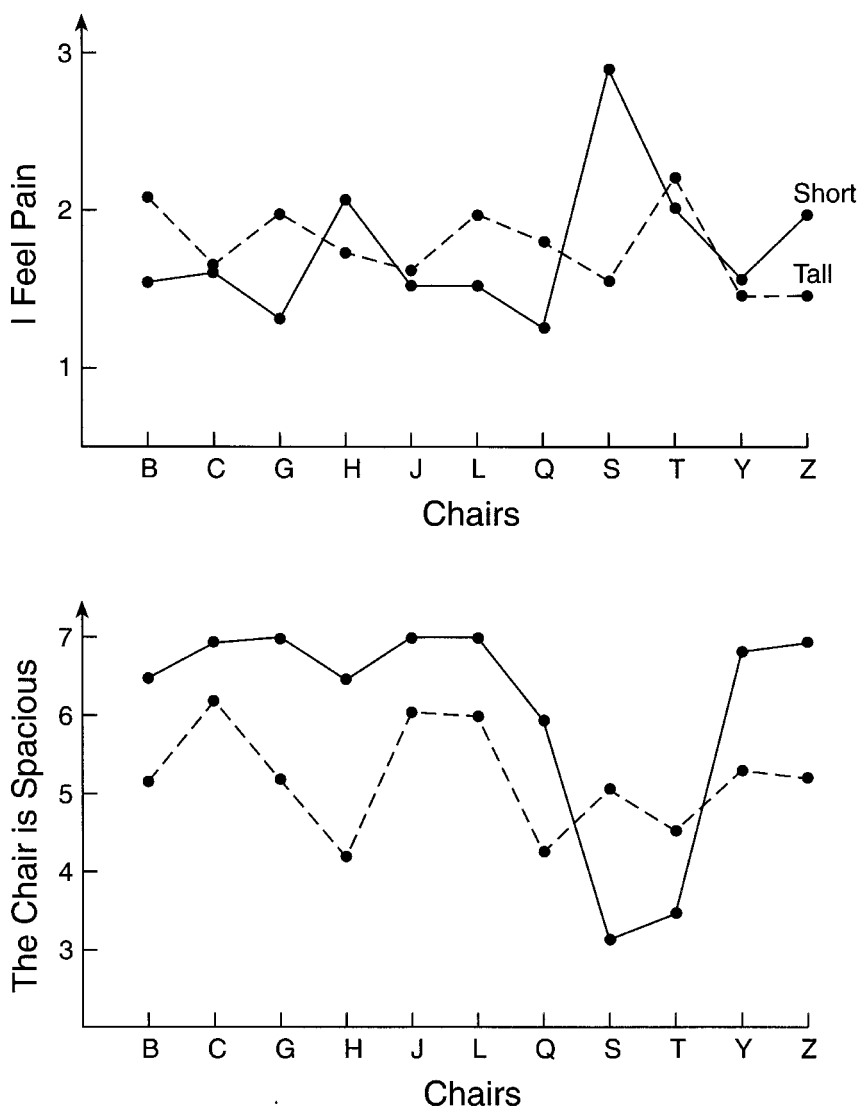


Figure 5. Significant Chair $\times$ Stature interactions for managers.

On the other hand short secretaries felt more tired (2.96) than tall secretaries (2.14) and also felt more cramped (1.91) than tall secretaries (1.57). For the comfort descriptors there were four significant main effects, and taller secretaries experienced greater comfort. Overall the significant differences were fairly small, about 0.7 scale units, and there may not be many practical implications.

Looking at the broader picture of the analysis, there are several important results. There was no significant Chair $\times$ Time period interaction. This implies that the rank order of the outcome of the testing did *not* change as a function of time of day. Clearly the chairs were not uniform in design. Some chairs were well designed and were highly rated and some were not so well designed and were rated lower. It

Table 6. Correlations of descriptors in the CEC with measures of overall comfort/discomfort, GCR, and BPD. For $|r| > 0.125$, $p < 0.01$ and for $|r| > 0.23$, $p < 0.001$. Values of $|r| > 0.4$ are underlined.

	Overall comfort	Overall discomfort	GCR	BPD
<i>Discomfort descriptors</i>				
I have sore muscles	-0.261	0.312	-0.367	<u>0.440</u>
I have heavy legs	-0.231	0.089	-0.102	0.126
I feel stiff	-0.362	<u>0.409</u>	<u>-0.502</u>	0.381
I feel restless	-0.324	<u>0.470</u>	<u>-0.489</u>	0.370
I feel tired	-0.028	0.176	-0.312	0.250
I feel painful	-0.329	0.352	<u>-0.402</u>	<u>0.440</u>
I feel numb	-0.138	0.164	-0.242	0.093
I feel uneven pressure	<u>-0.409</u>	<u>0.472</u>	-0.359	0.267
I feel cramped	-0.306	<u>0.455</u>	-0.209	0.099
<i>Comfort descriptors</i>				
The chair looks nice	<u>0.512</u>	-0.324	<u>0.406</u>	-0.168
The chair feels soft	<u>0.434</u>	-0.311	0.398	-0.187
The chair is spacious	<u>0.449</u>	-0.317	0.256	-0.100
I feel relaxed	<u>0.614</u>	-0.317	<u>0.556</u>	-0.278
I feel restless	<u>0.622</u>	<u>-0.449</u>	<u>0.530</u>	-0.249
I feel refreshed	<u>0.616</u>	<u>-0.483</u>	<u>0.521</u>	-0.229
I like the chair	<u>0.715</u>	<u>-0.500</u>	<u>0.601</u>	-0.339
GCR	<u>0.514</u>	<u>-0.431</u>		<u>-0.488</u>
BPD	-0.292	0.277		

had been expected that biomechanical problems (from uneven pressure, cut-off of circulation, stiffness, etc.) would develop over the day and that the problem would become greater for some chairs than for others. The results prove differently. Biomechanical discomfort factors increased as a function of time of day, and chair design did not seem to matter.

There are two kinds of discomfort descriptors: those related to biomechanics problems (Descriptors 1, 2, 3, and 5 in tables 4 and 5) and those related to chair design features (Descriptors 7 and 8). For the discomfort descriptors related to chair features: ‘uneven pressure’ and ‘cramped’ (items 7 and 8) there were significant differences between the chairs, but these descriptors did not change significantly over time. Thus, the discomfort resulting from ‘uneven seat pressure’ was immediately noticeable. Likewise, the significant chair effect for ‘cramped’, which is interpreted as the effect of constrained space (small seat) was also reported immediately.

There were two kinds of comfort descriptors: those related to ‘well-being’ (Descriptors 10–12) and those related to chair design (Descriptors 13–16). For all of them there were significant chair effects. There were no significant effects of ‘Time period’ or Chair × Time period interaction. These results indicate that there were indeed significant differences between chairs in terms of comfort, but that the rating of comfort did not change over time. Just as for the two discomfort chair descriptors (7 and 8) it would *not* be necessary to use a chair for one entire day to assess chair comfort. Rather, a person or a group of individuals testing a chair would be able to reliably assess comfort immediately.

For the overall evaluations at the bottom of tables 4 and 5 there were several significant results. However, since there were only 5 subjects in each category, results are more difficult to generalize and they are not discussed here.

3.3.3. Correlation analysis: Most of the descriptors in table 6 produced significant correlations with Overall comfort, Overall discomfort (Descriptors 17 and 18), GCR and BPD, indicating that the coefficients are statistically reliable. The coefficients in table 6 are not very large. This could be expected, since different descriptors contribute to different aspects of comfort, discomfort, GCR and BPD (Zhang *et al.* 1996).

All the correlation coefficients in table 6 had the expected signs, with comfort descriptors positively correlated with 'comfort' and negatively with 'discomfort' and vice versa. For GCR and BPD the signs were also consistent (a high value of GCR indicates comfort and a high value of BPD discomfort).

Owing to this consistent pattern one could possibly conclude that comfort and discomfort descriptors are just the opposites of each other and could be rated on one scale. However, the magnitude of the correlation coefficients indicate differently. The comfort descriptors had consistently greater correlations with 'overall comfort' than with 'overall discomfort'. In addition, the correlation coefficients between comfort descriptors and GCR (which is a composite measure of comfort/discomfort) were greater than with BPD (which measures discomfort). Similarly, the discomfort descriptors had greater correlations with 'overall discomfort' than with 'overall comfort'. This supports again the notion that comfort and discomfort are not just opposite extremes on the same continuum.

The results of the correlation analysis also indicate that GCR measured both comfort and discomfort. The slightly higher correlation with comfort descriptors than discomfort descriptors may be because the chairs in our sample were generally comfortable, and the tasks were not very demanding. Note that similar results were obtained in figure 3 where the factor for discomfort had a maximum score of 24, whereas the maximum score for comfort was 47.

3.4. Evaluation of revised checklist

Based on the results obtained in the first experiment the Chair Evaluation Checklist was revised. The 16 descriptors were reduced to 12, six for discomfort and six for comfort. 'I feel numb' was removed due to lack of significance in the analyses of variance (tables 4 and 5) and low correlation coefficients in table 6. In addition several subjects reported that they did not understand what was being referred to. 'I feel restful' was removed because responses were highly similar to 'I feel relaxed' (see also the results from the cluster analysis in Zhang *et al.* 1996). For similar reasons 'I feel pain' was eliminated because of high correlation with 'I have sore muscles' which is a more specific statement. The final checklist is presented in figure 6, which lists the 12 descriptors and two overall ratings of comfort and discomfort.

One possible advantage of this combined checklist is that ratings of discomfort descriptors precede the overall rating of discomfort and ratings of comfort descriptors precede the overall rating of comfort. The subjects can then consider the various factors involved which may help in defining the conceptual differences between comfort and discomfort, and the fact that they are not mere opposites. Despite these

DISCOMFORT factors are rated below										COMFORT factors are rated below									
	Not at all			Moderately			Extremely				Not at all			Moderately			Extremely		
	1	2	3	4	5	6	7	8	9		1	2	3	4	5	6	7	8	9
I have sore muscles										I feel relaxed									
I have heavy legs										I feel refreshed									
I feel uneven pressure from seat pan or seat back										The chair feels soft									
I feel stiff										The chair is spacious									
I feel restless										The chair looks nice									
I feel tired										I like the chair									
I feel uncomfortable										I feel comfortable									

Figure 6. The final Chair Evaluation Checklist. *Instructions to subjects:* Please rate the discomfort and comfort of this chair. Discomfort factors are listed to the left and comfort factors to the right. Mark an 'X' on each line at the point that best describes your feelings or impressions. Note: 1 = Not at all; 9 = extremely.

precautions, there was, however, still a possibility that subjects may simply rate comfort as the opposite of discomfort without reflecting on the different meanings.

The objectives of the second experiment were to: (1) validate the factor structure of comfort and discomfort using the modified checklist with fewer items; and (2) investigate possible transfer effects between the ratings of comfort and discomfort on the combined checklist.

4. Study 2

4.1. Method

4.1.1. *Subjects:* Thirty-seven office employees of SUNY at Buffalo participated in the experiment on a voluntary basis. Their ages ranged from 20 to 55 years, with a mean of 39 years and a SD of 8.6 years. A majority of the subjects were female and their office activities were predominantly typing, filing, and data processing.

4.1.2. *Checklists:* Three different checklists were used. Checklist A was a comfort checklist consisting of the seven items to the right in figure 6. This checklist was used as control. Checklist B was a discomfort checklist, consisting of the seven items to the left in figure 6. This checklist was also used as control. Checklist C used the entire checklist as presented in figure 6.

4.1.3. *Procedure:* Sitting comfort and discomfort were assessed using the three types of checklists during a time period of 3 weeks. The ratings were performed on Tuesday and Wednesday of each week, thereby leaving some time between conditions to minimize transfer effects. Subjects filled in the checklist three times per day; in the morning after coming to work, before the lunch break, and in the

afternoon before they went home. Subjects assessed their own working situation and chairs, and nothing was manipulated in the workplace.

To eliminate a bias due to sequence effects, subjects were randomly divided into two groups. One group used checklist A in week 1 and checklist B in week 2, while the order was reversed for the other group. Checklist C was used during the last week.

4.1.4. Evaluation method: Throughout the 6 days of evaluation each of the 37 subjects responded to 168 questions for a total of 6,216 data points. Two analyses were undertaken: Factor analysis and ANOVA. Factor analysis was performed to validate the multidimensional character of comfort and discomfort. Analysis of variance was performed to investigate the possible differences between the two procedures of presenting the checklist.

4.2. Results

Ratings of the descriptors in the CEC were submitted to Factor analysis with *varimax* rotation. Three factors were found (table 7). The first factor contained discomfort descriptors. The second factor pooled comfort descriptors related to chair design, and the third factor expressed comfort factors related to well-being. These three factors explained 79% of the variance.

A $2 \times 3 \times 2$ within-subjects ANOVA was performed for each of the descriptors in CEC. There were three fixed factors: Days (Tuesday and Wednesday), Time periods of day (morning, lunch and afternoon) and Checklist (A + B versus C). Subjects were treated as a random factor. The results are given in table 8.

4.3. Discussion

4.3.1. Factor analysis: The result of the factor analysis is consistent with the previous analyses. Three factors emerged explaining 79% of the variance. Factor 1 pooled all six discomfort descriptors. According to the first experiment this factor is not related to chair design, since discomfort increases over time but is largely unaffected by chair

Table 7. Results of factor analysis of descriptors of comfort and discomfort in the modified chair evaluation checklist.

	Factor 1 Discomfort	Factor 2 Comfort/chair design	Factor 3 Comfort/well-being
I feel stiff	0.901	-0.128	-0.161
I have sore muscles	0.880	-0.076	-0.044
I have heavy legs	0.874	-0.071	-0.135
I feel uneven pressure	0.686	-0.356	-0.118
I feel restless	0.683	-0.243	-0.317
I feel tired	0.688	-0.034	-0.530
Chair is spacious	-0.112	0.906	0.180
I like the chair	-0.150	0.887	0.114
Chair looks nice	-0.288	0.856	0.027
Chair feels soft	-0.038	0.810	0.304
I feel refreshed	-0.178	0.193	0.922
I feel relaxed	-0.241	0.265	0.853
Extracted variance	3.944	3.318	2.158

Table 8. Significant effect found in ANOVA in comparison of Checklist C with A+ B.

Descriptor	Day	Time period	Checklist	D ×T	D ×C	T ×C	D ×T ×C
1. I feel relaxed		0·0003					
2. I feel refreshed		0·0001					
3. Chair feels soft							
4. Chair is spacious							
5. Chair looks nice							
6. I like the chair							
7. I feel comfortable		0·028					
8. I have sore muscles							
9. I have heavy legs							
10. I feel uneven pressure		0·009	0·0001		0·044		
11. I feel stiff							
12. I feel restless		0·0001					
13. I feel tired		0·0004					
14. I feel comfortable		0·0002					

design features. In this study ‘Chairs’ was not an experimental factor, since the subjects used only one—their personal chair. The authors could not shed any further light on this issue therefore.

The second factor pooled all comfort descriptors related to chair design. This factor seems indeed, to be the only one related to chair design. The third factor ‘well-being’ had two major factor loadings: ‘I feel refreshed’ and ‘I feel relaxed’. There were also minor contributions from the factors ‘I feel tired’ and ‘Chair feels soft’. The results from this analysis supports the original theory as presented by Zhang *et al.* (1996). There are two aspects of sitting comfort: chair design and sense of well-being.

4.3.2. *Analyses of variance:* The results of the ANOVAs in table 8 show that, with one exception, there were significant effects only for time period. For the exception ‘I feel uneven pressure’ there was a significant interaction between days and checklists. Ratings in checklist C produced greater values than for Checklist A + B, and more so during the first 3 days than during the last 3 days. There is no obvious explanation for this, and considering the lack of significance for the other variables this result was considered to be less important.

For the chair factors ‘feels soft’, ‘spacious’, ‘looks nice’, and ‘I like the chair’ there were no significant effects of time periods. This is in agreement with the results obtained for the first study.

There were significant time period effects for the comfort descriptors ‘I feel relaxed’ and ‘I feel refreshed’, as well as for the overall evaluation of ‘I feel comfortable’. For three of the discomfort descriptors ‘I feel uneven pressure’, ‘I feel restless’, and ‘I feel tired’ there were significant time period effects. For three of descriptors ‘I have sore muscles’, ‘I have heavy legs’, and ‘I feel stiff’ there were no significant effects. This result was contrary to our expectations and the results of the first experiment. However, ANOVAs were also run for each type of checklist separately. In this case there were significant time period effects for all discomfort descriptors.

Overall the results showed that there was no significant effect of type of checklist on ratings of comfort and discomfort. Therefore comfort and discomfort can be rated at the same time using the checklist in figure 6. In fact, subjects verbally commented that they felt it was easy to estimate 'overall comfort' and 'overall discomfort' after being exposed to the descriptors, which conceptually clarified the procedure.

4.3.3. Discomfort and comfort: In the previous study 43 descriptors were classified and it has been proposed that comfort and discomfort are two independent factors that should be measured separately using different scales (Zhang *et al.* 1996). In this study the number of descriptors were reduced to 16 for the field evaluation, and 12 for the final validation. The reduction of the number of descriptors was undertaken to develop a composite checklist that would be practical to use for field evaluation of chairs, while portraying the various factors involved. Factor analyses for both experiments validated the theory although there were some minor differences between the analyses. 'Chair size' emerged as a third factor in the first analysis and 'well-being' in the second analysis. The first two factors 'comfort' and 'discomfort' were otherwise identical to the first study (Zhang *et al.* 1996). The second analysis performed on the final checklist, explained 79% of the variance. This was a significant increase from the first analysis, which explained 61% of the variance.

In the first field study it was found, not unexpectedly, that the main factor affecting discomfort is time period (time of the day). Time period correlates with fatigue, and therefore this result was expected (figure 1). More surprising was the result that although the evaluated chairs were clearly different and did produce significant differences for 'uneven pressure' and 'cramped', there were no significant 'Chair $\times$ Time' interactions. This implies that the rank order of preference among the set of chairs was established during the first assessment and did not change during the day. For the discomfort descriptors there were relatively few significant results related to 'chairs' (or chair design). This finding explains why previous researchers have had difficulties in differentiating between chairs in ergonomic evaluations. Shackel *et al.* (1969) evaluated 12 chairs but could only distinguish two groups with different comfort/discomfort. Hozeski and Rholes (1986) compared three seemingly very different chairs but obtained no significant differences. In two previous chair evaluations the present authors compared five different assessment methodologies for selecting the best chairs out of a sample of 20 chairs (Helander *et al.* 1987, Helander and Zhang 1991). The different methodologies produced different rank orders of the chairs, indicating that there was no consistent appreciation of discomfort/comfort. In addition the differences between chairs were generally small and only a few were significant.

One may question whether chair design is at all important for discomfort! Branton (1966, 1987) used an experimental chair that was designed to give minimal, but adequate, support of the ischial tuberosities and the lumbar. It had a very short seat pan (20 in. wide $\times$ 10 in. long) and a very small lumbar or back support (20 in. wide $\times$ 5 in. high), but no other support. Branton used it for one year in his office, and claimed that it was basically 'comfortable'.

Many researchers have observed that chair users, seemingly unaware, change posture constantly, supposedly to relieve the build up of body pressure and discomfort. For example, crossing legs and putting one leg on top of the other changes the pressure distribution under the ischial tuberosities. However, chair users are unaware of this behaviour, and it is not available for introspection. Postural

changes are normal and although they may be performed to reduce 'discomfort', the chair is not necessarily perceived as being uncomfortable.

It may seem that chair design and biomechanics of sitting are not so important—unless the chair user has a bad back (Michel and Helander 1994), or unless the design of the chair severely and obviously violates basic design criteria. Examples of such criteria are of a seat pan that is too long for small users, or a sharp seat edge that makes indentations in the legs, or a lumbar support that is either too high or too low for the user and disfigures the spine, or a seat cushion that is too thick or soft and generates a non-optimal seat pressure distribution (Gross *et al.* 1994), or a seat that does not allow the hip joint to open up to greater than 100°.

Most chairs conform with these design criteria, and it may not be surprising that it is difficult to rank order chairs by measuring discomfort. The human body is surprisingly adaptive and not very sensitive to variations in chair design. Ergonomic chairs may be 'over designed' and various claims to reduction of discomfort would probably not hold up in research—they are merely design gimmicks.

Design gimmicks may, however, affect comfort perception. Aesthetic design, plushness and softness are important elements of comfort and could provide convincing sales arguments for chair manufacturers. These types of design features are not functional requirements, but rather they become associated with a sense of well-being. All comfort descriptors produced significant differences between chairs (tables 4 and 5) and these variables were more reliable and more sensitive for evaluating and ordering chairs in terms of preference than the discomfort descriptors.

As long as basic design criteria are not violated, it may be difficult to justify discomfort assessment of chairs. Much of the discomfort that builds up during a day of sitting seems to be related to the passage of time rather than the design of chairs. For evaluation of office chairs without basic design flaws, the comfort descriptors are more informative than the discomfort descriptors. Comfort is in the end the more important factor, because there are significant differences between chairs which has clear design implications.

4.3.4. Chair evaluation checklist: In table 6 Shackel's General Comfort Rating (GCR) and Corlett's Body Part Discomfort (BPD) were compared to the sixteen descriptors used in the field evaluation. Not surprisingly BPD produced high correlations with discomfort descriptors, but the correlation with comfort descriptors was poor. This is to be expected, since the purpose of BPD is to provide a general tool for evaluation of discomfort (rather than comfort).

Since comfort is not included in BPD it is considered that BPD may be inappropriate for chair evaluations. From this perspective GCR is more appropriate, since it does include comfort descriptors. In addition, the correlation coefficients between the comfort descriptors and GCR were much greater than for BPD (table 6). However the authors' own metric for 'Overall comfort' produced greater correlation coefficients with the comfort descriptors than both GCR and BPD; this provides further validation of the Chair Evaluation Checklist.

There is one advantageous aspect of GCR that is very practical: it produces a single, bottom-line number. However, the use of a single scale combining comfort and discomfort is contrary to the present model and results.

With 12 different scales the CEC is more time consuming to use. The number of scales could have been reduced further to, say, six. However, one would lose some of the multifactorial nature of the assessment. The two 'bottom-line' measures, 'I feel

comfortable' and 'I feel uncomfortable' could be used in isolation. The authors feel, however, that it is necessary for subjects to go through the motions and evaluate all descriptors. The descriptors are informative and they help subjects to define comfort and discomfort.

4.3.5. Evaluation time: How long should subjects sit in a chair to evaluate it? Previous research has proposed that 2 h may be appropriate. In the evaluation in tables 4 and 5, there were no significant Chair $\times$ Time period interactions, and one can therefore conclude that the rankings obtained from the very first rating were satisfactory. In table 8 chairs were not compared, but the authors studied how the assessment of comfort and discomfort descriptors were affected by the time of the day that assessments were made. The comfort descriptors related to chair design (items 3–6) were unaffected by time, whereas comfort descriptors related to well-being (items 1–2) and most discomfort descriptors were affected by time. The authors believe that the time dependency simply illustrates fatigue effects, which is consistent with the model in figure 1.

It is difficult to propose a measurement procedure. One option is to only evaluate one chair per day at a standard time—e.g. between 09:00 and 11:00 h. Another method would be to evaluate all chairs in a sample during a very short time. The chairs could be located in one room and subjects would sit in each one for only a few minutes before filling in evaluation forms (Helander *et al.* 1987). The authors have no data to prove the validity of this method.

4.3.6. The model for perception of comfort and discomfort: The results from the plotting of factor loadings in figures 3–4 give further credence to the model in figure 1 for perception of comfort and discomfort (Zhang *et al.* 1996). The range of comfort ratings and factor scores were much greater for comfort than for discomfort. This contradicts some of the earlier definitions of comfort as being merely a binary state—comfort is absence of discomfort (Hertzbert 1972, Branton 1966). With an expanded definition of comfort, and using six descriptors, subjects had a greater range of evaluations available, and the average ratings of overall comfort were in fact much greater than the average ratings of overall discomfort. The authors believe that comfort and discomfort are independent factors that deserve differential measurement. The consistent results obtained in this field validation study together with the previous experimental study (Zhang *et al.* 1996), proves that both comfort and discomfort can be quantified independently. The scales developed for the Chair Evaluation Checklist seem to provide consistent results, which can be used for practical evaluations of chair comfort and discomfort.

In using the checklist it is necessary to consider the model in figure 1. Discomfort is affected by biomechanical factors and fatigue. For users of well-designed chairs fatigue (due to passage of time) is much more important than biomechanics.

Comfort assessment is affected by the aesthetics of chair design and a sense of well-being. The sense of well-being (feeling relaxed and refreshed) may also be affected by time. The results presented in tables 4 and 5 showed hardly any time dependence, whereas the results in table 8 did demonstrate a time dependence. These results are therefore somewhat inconsistent.

It is noted that there are similarities in the present model with Maslow's (1970) model for job satisfaction which is based on both physical needs (in the present model this would be discomfort) and well-being (comfort in the present model). It is noted that

in the present study comfort had two components that are usually associated with job satisfaction: 'feeling relaxed' and 'feeling refreshed'. In the future it is important to link research from different fields such as biomechanics, aesthetics design and job satisfaction. There are interesting commonalities, and they deserve to be explored.

The authors also note some similarities in their approach with that of Kansei Engineering, which uses expressions of customers' feelings as a guide to product design (Nagamachi 1995). In the present case the authors derived many expressions of feelings related to chair design, and these served the purpose of evaluating chairs. Kansei Engineering utilizes a data bank of 'feelings' that are associated with specific design features. For example, a steering wheel in a car may be associated with feelings such as 'solid', 'beautiful' and so forth. Considering the promising results in the present study, it would seem that one can expand the present approach and use expressions of feelings for the purpose of chair design.

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